

1. INTRODUCTION

1.1 Background of Study

Cervical cancer remains one of the leading causes of cancer-related mortality among women worldwide, particularly in developing countries. The progression of cervical cancer is characterized not only by persistent infection with high-risk human papillomavirus (HPV), but also by dysregulation of key cellular pathways that control apoptosis. Among these, the B-cell lymphoma 2 (Bcl-2) protein plays a crucial role as an anti-apoptotic regulator, promoting cancer cell survival by inhibiting programmed cell death. Overexpression of Bcl-2 has been widely associated with resistance to therapy and enhanced tumor progression in cervical cancer cells.[1]

In addition to host cellular mechanisms, viral oncoproteins such as HPV E6 contribute to carcinogenesis by interfering with tumor suppressor proteins like p53, further supporting uncontrolled cell proliferation. Therefore, targeting both apoptotic regulators such as Bcl-2 and viral proteins like HPV E6 represents a promising strategy for the development of effective anticancer therapies [2]

Conventional treatment approaches such as chemotherapy and radiotherapy are effective but often limited by side effects, high cost, and the emergence of resistance. This has led to increasing interest in identifying alternative therapeutic agents from natural sources. Medicinal plants, rich in bioactive phytochemicals, offer a promising approach due to their ability to target multiple pathways involved in cancer development. Many plant-derived compounds have shown the ability to induce apoptosis and inhibit tumor growth, making them valuable candidates in anticancer drug discovery [3]

1.2 Significance of *Acalypha indica*

Acalypha indica is an important medicinal plant belonging to the family Euphorbiaceae, genus *Acalypha*, and order Malpighiales. Commonly known as Indian Acalypha or Indian Nettle, it is also referred to by local names such as Kuppaimeni in Tamil and Haritamanjari in Sanskrit. The plant is widely distributed across tropical and subtropical regions and is native to India, where it

has been extensively used in traditional systems of medicine like Ayurveda and Siddha. It is commonly found in fields, roadsides, and wastelands, making it easily accessible and economically viable. Traditionally, *Acalypha indica* has been utilized for its antimicrobial and anti-inflammatory properties, as well as in the treatment of skin disorders, wound healing, and various respiratory and digestive ailments. In recent years, there has been growing scientific interest in this plant due to its rich phytochemical composition, including phenols, flavonoids, and alkaloids, which are known to exhibit significant biological activities. Importantly, the exploration of native medicinal plants such as *Acalypha indica* offers considerable advantages in drug discovery, as they are readily available, cost-effective, and generally associated with fewer severe side effects. Therefore, investigating its anticancer potential not only helps bridge traditional knowledge with modern scientific research but also contributes to the development of sustainable and accessible therapeutic strategies.[4-7]

1.3 Objective

The present study aims to investigate the anticancer potential of *Acalypha indica* against cervical cancer using an integrated experimental and computational approach. The specific objectives include:

- To perform qualitative phytochemical screening of *Acalypha indica* extract
- To identify bioactive compounds using GC–MS analysis
- To evaluate molecular interactions of selected compounds with cancer-related proteins (HPV E6, Bcl-2) using SwissDock
- To assess pharmacokinetic properties through ADME analysis
- To determine cytotoxic effects on HeLa cells using MTT assay
- To evaluate apoptosis through DNA fragmentation analysis

2. REVIEW OF LITERATURE

2.1 Phytochemical Studies of Medicinal Plants

Medicinal plants have long been recognized as a valuable source of therapeutic compounds. They contain a wide variety of secondary metabolites such as alkaloids, flavonoids, tannins, and terpenoids, which are responsible for their biological activities. These compounds play a crucial role in protecting plants from environmental stress and pathogens, and they also exhibit significant pharmacological properties in humans. Phytochemical screening is therefore an essential step in identifying these compounds and understanding their potential medicinal value.[8]

2.2 *Acalypha indica* in Anticancer Research

Previous studies on *Acalypha indica* have primarily focused on its antimicrobial and antioxidant activities. However, recent research has started exploring its anticancer potential. Some studies suggest that extracts of the plant can inhibit the growth of cancer cells and induce apoptosis. Although these findings are promising, more detailed studies are required to fully understand its mechanism of action and effectiveness against specific cancers such as cervical cancer.[9,10]

2.3 GC–MS Analysis in Drug Discovery

GC–MS is a powerful analytical technique widely used for identifying chemical compounds in plant extracts. It separates compounds based on their volatility and provides detailed information about their molecular structure. In drug discovery, GC–MS plays a crucial role in identifying potential bioactive compounds that can be further studied for therapeutic applications. This technique helps bridge the gap between traditional knowledge and modern pharmacological research.[11]

2.4 Molecular Docking in Cancer Research

Molecular docking is an important computational method used to predict how small molecules (ligands) interact with target proteins. It helps in understanding the binding affinity and stability

of these interactions, which are critical for drug development. In cancer research, docking studies are often used to identify compounds that can interact with proteins involved in apoptosis and cell survival pathways.[12,13]

2.5 ADME and Drug-Likeness Studies

ADME analysis evaluates how a compound behaves inside the body, including its absorption, distribution, metabolism, and excretion. These properties are essential for determining whether a compound can be developed into a drug. Drug-likeness studies also help identify compounds that meet criteria such as Lipinski's rule of five, ensuring better bioavailability and effectiveness.[14]

2.6 Summary of Previous Studies

Overall, previous studies emphasize the importance of combining phytochemical analysis, computational modeling, and biological assays. This integrated approach enhances the identification and validation of potential anticancer agents derived from natural sources.

3. MATERIALS AND METHODS

3.1 Materials Used

- Fresh leaves of *Acalypha indica* collected from Tiruvannamalai, Tamil Nadu, India
- Distilled water, ethanol, chloroform, methanol
- Hydrochloric acid (HCl), sulfuric acid (H₂SO₄), sodium hydroxide (NaOH), ferric chloride (FeCl₃), sodium chloride (NaCl), acetic acid
- Wagner's reagent
- Dimethyl sulfoxide (DMSO)
- GC–MS system with DB-5 column
- SwissDock and SwissADME online tools
- Human cervical cancer cell line (HeLa)
- Dulbecco's Modified Eagle's Medium (DMEM), fetal bovine serum (FBS)
- Trypsin–EDTA solution
- MTT reagent

- Phosphate-buffered saline (PBS)
- DNA isolation reagents (spin column-based method)
- Agarose, ethidium bromide, TAE buffer
- 96-well plates, micropipettes, centrifuge tubes

All chemicals and reagents used were of analytical grade.

3.2 Collection and Preparation of Plant Extract

Fresh leaves of *Acalypha indica* were collected and washed thoroughly with distilled water to remove impurities. The leaves were shade-dried at room temperature and ground into a fine powder.

An aqueous extract was prepared by heating the powdered material in distilled water using a heating mantle, followed by cooling and filtration. An ethanolic extract was prepared separately by macerating the powdered sample in ethanol for a suitable duration, followed by filtration to obtain the crude extract.



Fig.1: Shade drying, Powder preparation and extract preparation from *Acalypha indica* leaves

3.3 Qualitative Phytochemical Analysis

Qualitative phytochemical screening was performed to identify the presence of bioactive compounds.

3.3.1 Test for Alkaloids

Dried extract (0.1 g) was dissolved in ethanol and filtered. The filtrate was acidified using hydrochloric acid and treated with Wagner's reagent. The formation of a reddish-brown precipitate indicated alkaloids.

3.3.2 Test for Saponins

Extract was mixed with distilled water and vortexed. The formation of stable foam indicated saponins.

3.3.3 Test for Flavonoids

Ethanol extract was treated with sodium chloride and hydrochloric acid. A color change indicated flavonoids.

3.3.4 Test for Tannins

Aqueous extract was treated with ferric chloride. A blue-black or greenish coloration indicated tannins.

3.3.5 Test for Terpenoids (Salkowski Test)

Extract was mixed with chloroform and concentrated sulfuric acid. A reddish-brown interface indicated terpenoids.

3.3.6 Test for Quinones

Extract was treated with sodium hydroxide solution. Formation of a colored solution indicated quinones.

3.3.7 Test for Cardiac Glycosides (Keller–Killiani Test)

Extract was treated with chloroform, followed by acetic acid containing ferric chloride and concentrated sulfuric acid. Formation of a brown ring indicated cardiac glycosides. [15,16]

3.4 GC–MS Analysis

GC–MS analysis was performed to identify phytochemical constituents present in the *Acalypha indica* extract. The analysis was carried out using a gas chromatograph coupled with a mass spectrometer equipped with a DB-5 capillary column. The oven temperature was programmed from 100 °C to 270 °C at a rate of 10 °C/min. Helium was used as the carrier gas at a constant flow rate of 1.2 mL/min. The mass spectrometer operated in scan mode. Compounds were identified based on their retention time and comparison with standard mass spectral libraries [17,18]

3.5 Molecular Docking Using SwissDock

Molecular docking studies were performed to evaluate the interaction between selected phytocompounds and target proteins associated with cervical cancer.

The three-dimensional structures of Bcl-2 and HPV E6 proteins were obtained from the Protein Data Bank. Protein structures were prepared by removing heteroatoms and selecting the appropriate chain. Ligand structures (2,4-di-tert-butylphenol, tetradecanal, and pyridine-3-carboxamide) were retrieved from the PubChem database. Docking was carried out using the SwissDock online server under default parameters. Binding affinities and interaction patterns were analyzed based on docking scores [19,20]

3.6 ADME Analysis

ADME properties of selected compounds were evaluated using the SwissADME online tool. Ligand structures obtained from PubChem were uploaded to the server. Parameters such as absorption, distribution, metabolism, excretion, and drug-likeness were analyzed to assess their suitability as potential drug candidates.[21]

3.7 Cell Culture (HeLa Cells)

HeLa cells were cultured in DMEM supplemented with 10% fetal bovine serum and maintained at 37 °C in a humidified incubator with 5% CO₂. Cells were subcultured using trypsin–EDTA and used for further experiments.

3.8 Cytotoxicity Assay (MTT Assay)

HeLa cells were seeded into 96-well plates at a density of approximately 1×10^4 cells per well and incubated for 24 h. Cells were treated with different concentrations of *Acalypha indica* extract, while untreated cells served as control. After 24 h, MTT solution was added and incubated for 4 h. Formazan crystals formed were dissolved in DMSO, and absorbance was measured at 570 nm. Cell viability (%) was calculated using: [22]

$$\text{Cell viability (\%)} = \frac{\text{Sample OD}}{\text{Control OD}} \times 100$$

3.9 DNA Fragmentation Analysis

HeLa cells were treated with extract at IC₅₀ concentration and incubated for 24 h. Cells were harvested, washed with PBS, and subjected to DNA isolation. Extracted DNA was analyzed using 1% agarose gel electrophoresis containing ethidium bromide. DNA ladder was used as a marker. DNA bands were visualized under UV light, and the presence of a ladder pattern indicated apoptosis.[23]

4. RESULT AND DISCUSSION

The present study evaluated the anticancer potential of *Acalypha indica* through a combination of phytochemical screening, GC–MS analysis, molecular docking, ADME prediction, cytotoxicity assay, and DNA fragmentation analysis. The results collectively provide strong evidence supporting the biological activity of the plant extract.

4.1 Phytochemical Screening

Qualitative phytochemical analysis revealed the presence of several important secondary metabolites (Table 4.1). Alkaloids, flavonoids, tannins, and quinones were found to be strongly present (+++), while terpenoids and cardiac glycosides showed moderate presence (++). Saponins were not detected (Fig.1).

Table 4.1: Qualitative Phytochemical Screening of *Acalypha indica*

S. No	Phytochemical	Result
1	Alkaloids	+++
2	Saponins	–
3	Flavonoids	+++
4	Tannins	+++
5	Terpenoids	++
6	Quinones	+++
7	Cardiac glycosides	++

Note: (–) Not detected | (+) Mild | (++) Moderate | (+++) Strong

These phytochemicals are known for their antioxidant, anti-inflammatory, and anticancer properties, suggesting that the plant extract possesses significant therapeutic potential [4,9]

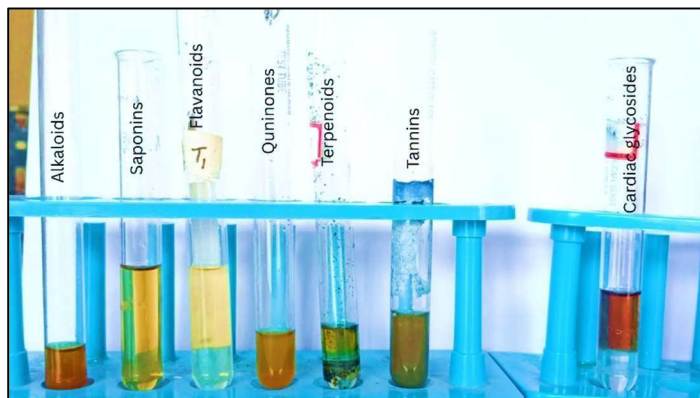


Fig.2: Qualitative phytochemical analysis of *Acalypha indica*

4.2 GC–MS Analysis

GC–MS analysis confirmed the presence of multiple phytoconstituents in the extract (Fig.2). Based on their relevance and potential biological activity, three compounds were selected for further study (Table 4.2).

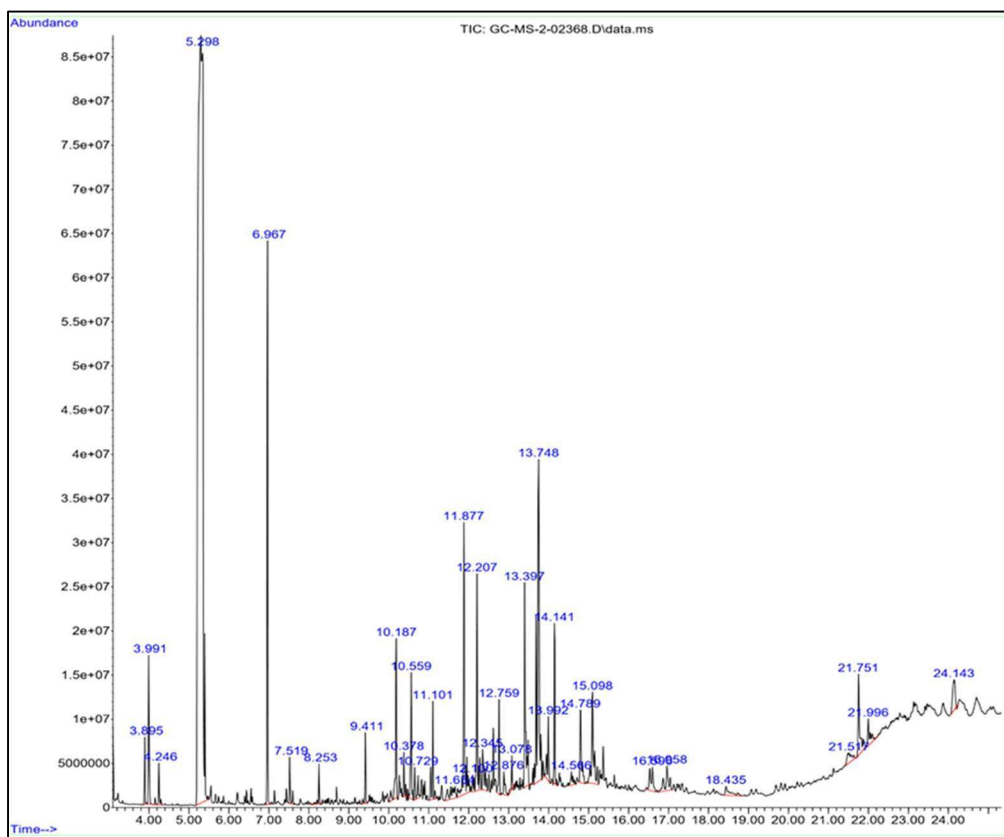


Fig.3: GC-MS analysis of *Acalypha indica*

Table 4.2: Major Compounds Identified by GC–MS

S.no	Retention Time (min)	Compound Name	Molecular Formula	Molecular Weight	Peak Area (%)
1	5.298	Phenol	C ₆ H ₆ O	94.11	50.19
2	6.967	Acetic acid, 2-ethylhexyl ester	C ₁₀ H ₂₀ O ₂	172.26	5.22
3	10.378	2,4-di-tert-butylphenol	C ₁₄ H ₂₂ O	206.32	0.49
4	11.877	Hexadecane, 2,6,10,14-tetramethyl-	C ₂₀ H ₄₂	282.55	3.98
5	12.100	Pyridine-3-carboxamide, oxime	C ₆ H ₇ N ₃ O	137.14	0.31
6	12.207	Hexadecane, 2-methyl-	C ₁₇ H ₃₆	240.47	2.77
7	12.759	Tetradecanal	C ₁₄ H ₂₈ O	212.37	1.00
8	13.078	1,2-Benzenedicarboxylic acid	C ₈ H ₆ O ₄	166.13	0.34

Among these, 2,4-di-tert-butylphenol is reported to exhibit antioxidant and cytotoxic activities, making it a promising candidate for anticancer evaluation.[24,25]

4.3 Molecular Docking Analysis

Molecular docking was performed using Bcl-2 and HPV E6 proteins and 2,4-Di-tertbutylphenol, Tetradecanal, Pyrimidine-3-Carboxamide as ligands to evaluate ligand–protein interactions. The docking results (Table 4.3) showed that all compounds interacted with the target proteins, but with varying binding affinities.

Table 4.3: Molecular Docking Results

S. No	Ligand	Target Protein	Binding Affinity (kcal/mol)
1	2,4-di-tert-butylphenol	HPV E6	-5.423
2	2,4-di-tert-butylphenol	Bcl-2	-6.010
3	Tetradecanal	HPV E6	-4.187
4	Tetradecanal	Bcl-2	-4.772
5	Pyridine-3-carboxamide	HPV E6	-4.236
6	Pyridine-3-carboxamide	Bcl-2	-4.711

The highest binding affinity was observed for 2,4-di-tert-butylphenol with Bcl-2 (−6.01 kcal/mol), suggesting a relatively stable interaction (Fig.3). Since Bcl-2 plays a crucial role in cervical cancer progression by helping cells avoid apoptosis (programmed cell death).[26]

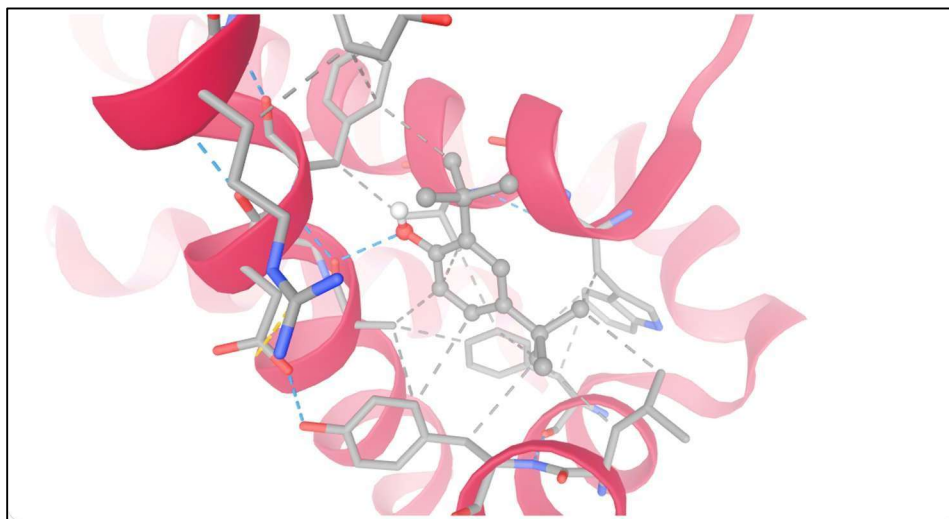


Fig.4: Docking position of ligand (2,4-Di-tert-butylphenol) – Protein (Bcl-2) Interaction

4.4 ADME Analysis

The pharmacokinetic properties of the selected compounds were evaluated using ADME analysis (Table 4.4)

Table 4.4: ADME Properties of Selected Compounds

Property	2,4-di-tertbutylphenol	Tetradecanal	Pyrimidine-3-carboxamide
Lipinski Rule	Yes; 0 violations	Yes; 0 violations	Yes; 0 violations
GI Absorption	High	High	High
Bioavailability Score	0.55	0.55	0.55
Drug-likeness	High	Moderate	Moderate

Among the compounds, 2,4-di-tert-butylphenol exhibited the most favourable drug-likeness, indicating its suitability as a potential drug candidate.

4.5 Cytotoxicity Assay (MTT Assay)

The cytotoxic effect of *Acalypha indica* extract on HeLa cells was evaluated using the MTT assay. A dose-dependent decrease in cell viability was observed, where increasing concentrations of the extract resulted in reduced cell survival (Table 4.5).[27]

Table 4.5: Effect of *Acalypha indica* Extract on HeLa Cell Viability

Concentration ($\mu\text{g/mL}$)	Cell Viability (%)	OD
Control	100	0.093
15.62	89.2	0.083
31.25	74.1	0.069
62.5	59.8	0.056
125	50.1	0.047
150	32.95	0.031

Approximately 50% cell viability (IC_{50}) was observed at a moderate concentration, indicating significant cytotoxic activity (Fig.4). The reduction in viability is associated with decreased mitochondrial activity, as reflected by reduced formazan formation

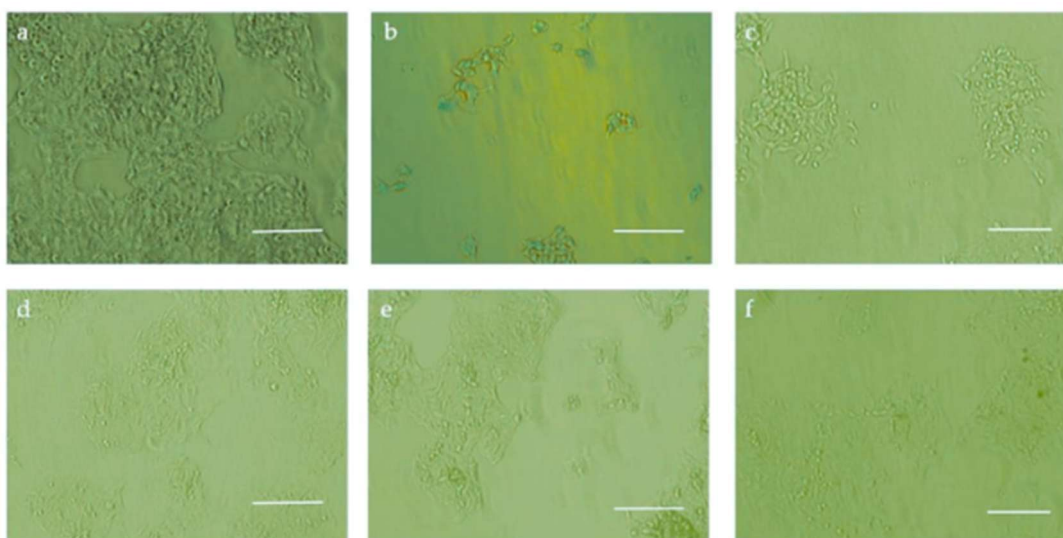


Fig.5: Cytotoxicity analysis (MTT assay) against HeLa cells at different concentration with untreated control

4.6 DNA Fragmentation Analysis

DNA fragmentation analysis demonstrated a distinct ladder-like pattern in treated cells, whereas untreated control cells showed intact genomic DNA. This fragmentation is a hallmark of apoptosis, indicating that the extract induces programmed cell death in cervical cancer cells.

The results strongly correlate with the cytotoxicity findings, confirming that the reduction in cell viability is primarily due to apoptotic mechanisms rather than necrosis.[28,29]

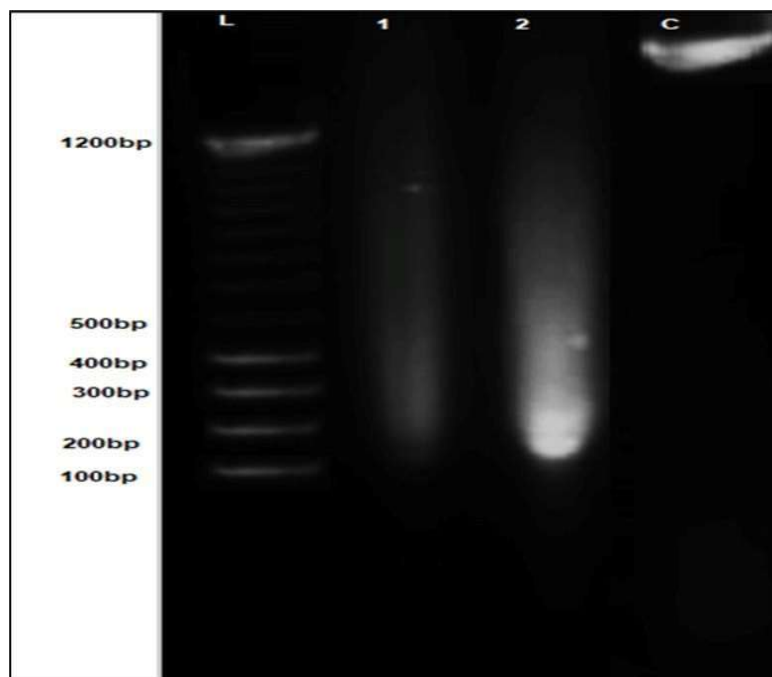


Fig.6: DNA fragmentation analysis of HeLa cells treated with *Acalypha indica*

5. CONCLUSION

The present study successfully demonstrated the anticancer potential of *Acalypha indica* through an integrated approach combining phytochemical analysis, GC–MS profiling, molecular docking, ADME evaluation, cytotoxicity assay, and DNA fragmentation analysis.

Qualitative phytochemical screening confirmed the presence of key bioactive compounds such as alkaloids, flavonoids, tannins, terpenoids, quinones, and cardiac glycosides, indicating the rich medicinal value of the plant. GC–MS analysis further identified several phytoconstituents, among which 2,4-di-tert-butylphenol, tetradecanal, and pyridine-3-carboxamide were selected for computational studies based on their structural and biological relevance.

Molecular docking analysis revealed that 2,4-di-tert-butylphenol exhibited the highest binding affinity with the Bcl-2 protein, suggesting its potential to interfere with viral oncogenic activity associated with cervical cancer progression. ADME analysis supported this finding by indicating favorable drug-likeness properties for the same compound.

The biological assays reinforced the computational results, as the *Acalypha indica* extract demonstrated a clear dose-dependent cytotoxic effect against HeLa cervical cancer cells. Furthermore, DNA fragmentation analysis confirmed the induction of apoptosis, indicating that the observed cytotoxicity is primarily mediated through programmed cell death mechanisms.

Overall, the findings suggest that *Acalypha indica* possesses significant anticancer potential and may serve as a promising source of bioactive compounds for the development of plant-based therapeutics against cervical cancer. However, further in-depth studies, including isolation of active compounds, in vivo validation, and mechanistic investigations, are required to fully establish its therapeutic efficacy.

6. SCOPE FOR FUTURE WORKS

Future studies should focus on the isolation and structural characterization of active compounds from *Acalypha indica* using advanced techniques such as NMR and LC–MS. Further in vitro and in vivo studies are required to validate anticancer activity, along with detailed mechanistic investigations targeting apoptosis-related pathways, particularly Bcl-2. Additionally, approaches like nanoformulation and combination therapy may be explored to enhance therapeutic efficacy.

In addition, detailed mechanistic studies targeting apoptosis-related pathways, particularly Bcl-2 regulation, should be carried out to better understand the mode of action. The development of nanoformulations and exploration of combination therapies may further enhance the bioavailability and therapeutic potential of the identified compounds

7. REFERENCES

- [1] D. M. Parkin, F. Bray, J. Ferlay, and P. Pisani, "Global cancer statistics, 2002," *CA: A Cancer Journal for Clinicians*, vol. 55, no. 2, pp. 74–108, 2005.
- [2] M. Schiffman, P. E. Castle, J. Jeronimo, A. C. Rodriguez, and S. Wacholder, "Human papillomavirus and cervical cancer," *The Lancet*, vol. 370, no. 9590, pp. 890–907, 2007.
- [3] A. Reyaz, S. S. Qadri, D. Javaid, S. Y. Ganie, and M. S. Reshi, "Exploring the role of traditional medicinal plants in cancer therapy: present efficacy and future directions," *Phytomedicine Plus*, vol. 5, no. 3, p. 100845, 2025.
- [4] R. K. Gupta, A. Sharma, and M. Sharma, "Phytochemical and pharmacological properties of *Acalypha indica*: A review," *International Journal of Pharmaceutical Sciences Review and Research*, vol. 20, no. 2, pp. 35–40, 2013.
- [5] S. Kumar, V. Kumar, and O. Prakash, "Pharmacognostic and phytochemical investigation of *Acalypha indica* Linn.," *Asian Pacific Journal of Tropical Biomedicine*, vol. 2, no. 3, pp. S190–S193, 2012.
- [6] P. S. Devi, R. S. Kumar, and M. S. Kumar, "Evaluation of antioxidant and anticancer activity of *Acalypha indica* extracts," *Journal of Applied Pharmaceutical Sciences*, vol. 1, no. 8, pp. 74–77, 2011.
- [7] V. S. Kumar, K. Navaratnam, and S. S. Selvaraj, "Antimicrobial activity of *Acalypha indica* Linn.," *International Journal of Pharma and Bio Sciences*, vol. 1, no. 3, pp. 1–6, 2010.
- [8] R. N. S. Yadav and M. Agarwala, "Phytochemical analysis of some medicinal plants," *Journal of Phytology*, vol. 3, no. 12, Dec. 2011.
- [9] S. Govindarajan, M. Vijayakumar, and A. R. Pushpangadan, "Antioxidant and anticancer activities of *Acalypha indica* Linn. leaf extract," *International Journal of Pharma and Bio Sciences*, vol. 2, no. 2, pp. 1–10, 2011.
- [10] K. K. S. Kumar, P. S. Babu, and R. R. Rao, "In vitro cytotoxic activity of *Acalypha indica* leaf extract against human cancer cell lines," *Journal of Pharmacy Research*, vol. 4, no. 6, pp. 2058–2060, 2011.
- [11] A. K. Verma, S. Patel, R. Singh, and P. K. Sharma, "GC–MS analysis, ADME prediction, antioxidant and anti-hyperglycemic activity of plant extracts," *Frontiers in Chemistry*, vol. 12, Art. no. 1458505, 2024.
- [12] G. Nigussie, A. Nigussie, R. Abebe, B. Tesfaye, and M. T. Gebrehiwot, "Phytochemical profiling, cytotoxicity, RT-PCR and molecular docking studies of plant-derived compounds,"

[13] G. Nigussie, A. Nigussie, R. Abebe, B. Tesfaye, and M. T. Gebrehiwot, "GC–MS metabolite profiling and multi-target docking analysis of plant extracts for cytotoxic and antioxidant activity," *Scientific Reports*, vol. 15, Art. no. 32835, 2025.

[14] Z. Abbaoui et al., "Synthesis, characterization, and anticancer evaluation of N-heterocyclic entities: ADME profiling and in silico predictions," *Toxicology Reports*, vol. 16, p. 102184, 2026.

[15] J. B. Harborne, *Phytochemical Methods: A Guide to Modern Techniques of Plant Analysis*, 3rd ed. London, U.K.: Chapman & Hall, 1998.

[16] G. E. Trease and W. C. Evans, *Pharmacognosy*, 16th ed. London, U.K.: Saunders Elsevier, 2009.

[17] R. P. Adams, *Identification of Essential Oil Components by Gas Chromatography/Mass Spectrometry*, 4th ed. Carol Stream, IL, USA: Allured Publishing, 2007.

[18] D. A. Skoog, F. J. Holler, and S. R. Crouch, *Principles of Instrumental Analysis*, 6th ed. Belmont, CA, USA: Thomson Brooks/Cole, 2007.

[19] M. Bugnon, U. F. Röhrig, M. Goullieux, M. A. S. Perez, A. Daina, O. Michielin, and V. Zoete, "SwissDock 2024: Major enhancements for small-molecule docking with Attracting Cavities and AutoDock Vina," *Nucleic Acids Research*, 2024.

[20] J. Eberhardt, D. Santos-Martins, A. F. Tillack, and S. Forli, "AutoDock Vina 1.2.0: New docking methods, expanded force field, and Python bindings," *Journal of Chemical Information and Modeling*, vol. 61, no. 8, pp. 3891–3898, 2021.

[21] A. Daina, O. Michielin, and V. Zoete, "SwissADME: A free web tool to evaluate pharmacokinetics, drug-likeness and medicinal chemistry friendliness of small molecules," *Scientific Reports*, vol. 7, p. 42717, 2017.

[22] M. Ghasemi, T. Turnbull, S. Sebastian, and I. Kempson, "The MTT assay: Utility, limitations, pitfalls, and interpretation in bulk and single-cell analysis," *International Journal of Molecular Sciences*, vol. 22, no. 23, p. 12827, 2021.

[23] S. Kasibhatla, G. P. Amarante-Mendes, D. Finucane, T. Brunner, E. Bossy-Wetzel, and D. R. Green, "Analysis of DNA fragmentation using agarose gel electrophoresis," *Cold Spring Harbor Protocols*, vol. 2006, no. 1, p. pdb-prot4429, 2006.

- [24] S. Kalaivani, S. Vidhya, and K. S. Janakiraman, "GC–MS analysis of phytochemical constituents in ethanolic leaf extract of medicinal plant," *International Journal of Pharmaceutical Sciences and Research*, vol. 6, no. 3, pp. 987–992, 2015.
- [25] P. Arora and S. Kaur, "GC–MS analysis of bioactive compounds in medicinal plants and their biological activities," *Journal of Pharmacognosy and Phytochemistry*, vol. 6, no. 4, pp. 190–195, 2017.
- [26] G. Chen and X. Deng, "Targeting Bcl-2 in cancer," *Oncoscience*, vol. 2, no. 12, pp. 883–884, 2015. DOI: 10.18632/oncoscience.232
- [27] T. L. Riss, R. A. Moravec, A. L. Niles, S. Duellman, H. A. Benink, T. J. Worzella, and L. Minor, "Cell viability assays," in *Assay Guidance Manual*, G. S. Sittampalam et al., Eds. Bethesda, MD, USA: Eli Lilly & Company and the National Center for Advancing Translational Sciences, 2016.
- [28] X. Liu, C. Kim, J. Yang, R. Jemmerson, and X. Wang, "Induction of apoptotic program in cell-free extracts: requirement for dATP and cytochrome c," *Cell*, vol. 86, no. 1, pp. 147–157, 1996.
- [29] S. Fulda and K. M. Debatin, "Extrinsic versus intrinsic apoptosis pathways in anticancer chemotherapy," *Oncogene*, vol. 25, no. 34, pp. 4798–4811, 2006.